

CANDIDATE CASE STUDY

Design Your Own Production-Ready Full-Stack GenAI Solution

YOUR CHOICE Business use case	REQUIRED Engineering capabilities	REQUIRED Production readiness
Choose and justify a realistic problem	Full stack, secure REST APIs and GenAI	End to end full stack application with monitoring

SUBMISSION TIMELINE Deliverables are expected within 48 hours. Choose a scope that allows you to demonstrate depth, sound trade-offs and a complete working journey.

1. Objective

Design and implement a production-minded full-stack application that uses Generative AI to solve a realistic business problem of your choice. You are expected to identify the problem, define the target users, establish the business value, select an appropriate GenAI workflow, and deliver a secure working solution.

You must propose your own use case The interview team will not prescribe the domain or business workflow. Selecting, narrowing and justifying the use case is part of the assessment.

2. What We Are Assessing

Candidates must submit a working repository that includes the backend service, frontend application, Docker configuration, setup instructions, sample environment file and a short technical explanation of the architecture, trade-offs, and production-readiness decisions.

Area	What should be demonstrated
Problem framing	Clear users, pain point, business value, success criteria and appropriate scope.
Full-stack engineering	A coherent frontend and backend with maintainable boundaries and a functional end-to-end journey.
Security	Authentication, authorization, safe API design, secure configuration and protection of sensitive information.
GenAI engineering	Strict structured outputs, prompt separation, injection resistance, deterministic boundaries and resilient provider integration.
Production readiness	Docker, structured logging, LLMOps
Communication	Ability to explain trade-offs, limitations, omitted work and the path to production.

3. Use Case Selection - CANDIDATE OWNED

Choose a use case where GenAI adds clear value . The solution should contain at least one GenAI operation that produces structured information or a controlled decision-support output used by the application.

3.1 Your Proposal Must Define

- The business problem and why it matters.
- The primary users and their roles.
- The core end-to-end user journey.
- Why GenAI is appropriate and where deterministic code remains necessary.
- The data being handled and its sensitivity.

3.2 Illustrative Domains

Examples include customer service, compliance, procurement, finance operations, insurance, knowledge management, human resources, project delivery, document intelligence, risk analysis or another professional-services use case. These are examples only. Original and well-justified choices are welcome.

4. Technical Constraints - NON-NEGOTIABLE

Layer	Required technology
Backend	Python 3.11+ and FastAPI
Validation	Pydantic v2
GenAI SDK	Native openai SDK or google-genai SDK; no LangChain or LlamaIndex
Persistence	SQLAlchemy or SQLModel with SQLite or PostgreSQL
Frontend	React or Next.js with TypeScript
Containerization	Backend Dockerfile, frontend Dockerfile and Docker Compose
Configuration	Environment variables, .env.example and no committed secrets

5. Authentication and Authorization - REQUIRED

Enterprise identity discussion The implementation may use local authentication, but the candidate must explain how the design would integrate with an enterprise identity provider using OAuth 2.0 or OpenID Connect.

6. API Security - REQUIRED

7. GenAI Engineering - REQUIRED

8. LLMOps - REQUIRED

9. Docker and Local Execution - REQUIRED

10. Scope Guidance

Depth over breadth You are not expected to create a complete enterprise product . A focused end-to-end workflow with strong security, observability and clearly explained trade-offs is preferable to a large but fragile feature set.

If an item is designed but not fully implemented, clearly identify it, provide the intended production design, and explain why it was deprioritized.

